

# Arduino - Smart Obstacle-Avoiding Rover

Explore Robotics, Ultrasonic Sensors & Autonomous Logic with Your Mini Robot!

## 1. How Does a Robot See & Decide?

Just like Chandrayaan's **Pragyan rover** navigates lunar terrain without a human driver, an autonomous rover uses sensors and software logic to detect obstacles and avoid collisions.

### **Ultrasonic Sensor (HC-SR04) — The Bat's Eyes**

Emits high-frequency sound waves and measures how long the echo takes to bounce back, working like bats in the Western Ghats!

### **Arduino UNO — The Microcontroller Brain**

Processes incoming sensor data instantly and determines rover movement, acting like traffic police at a busy crossroads.

## 2. Hardware Components & Wiring Callouts

### Core Hardware Setup

- **Arduino UNO:** Main logic processing board (The Brain).
- **HC-SR04 Ultrasonic Sensor:** Distance measurement unit (The Eyes).
- **L298N Motor Driver:** Power amplifier controlling motor spin and speed (The Accelerator).

| Component / Pin           | Target Pin              | Function / Purpose                                       |
|---------------------------|-------------------------|----------------------------------------------------------|
| Ultrasonic Trig Pin       | Arduino Pin 9           | Outputs sound wave pulses to trigger distance detection. |
| Ultrasonic Echo Pin       | Arduino Pin 10          | Listens for reflected sound wave echo signal.            |
| Motor Driver Control Pins | Arduino Pins 5, 6, 7, 8 | Controls directional rotation and speed of wheel motors. |

### 3. Distance Calculation & Decision Logic

The ultrasonic sensor operates by transmitting a sound pulse and timing its reflection—similar to echoing your voice in a Rajasthan haveli!

#### Distance Formula

$$\text{distance} = (\text{pulseDuration} \times \text{speedOfSound}) / 2$$

\* Note: The total time is divided by 2 because the sound wave travels to the obstacle and bounces back.

#### Pseudocode & Obstacle Decision Algorithm

```
// Send pulse and measure return echo duration pulseDuration = pulseIn(echoPin, HIGH); distance = (pulseDuration * 0.0343) / 2; if (distance < 15) { // Obstacle detected within threshold! stopMoving(); turnRight(); } else { // Path is clear! moveForward(); }
```

#### Threshold Distance Analysis

- **Too Short (2 cm):** The rover moves too fast and collides with the wall before it has time to brake!
- **Just Right (15 cm):** Optimal safe stopping distance to evaluate surroundings and turn.
- **Too Long (50 cm):** The rover stops prematurely in open areas long before reaching objects.

### 4. Testing & Troubleshooting Checklist

#### Motors Running Backward?

Simply switch the physical terminal wire connections on the motor driver, or flip output pin logic in your code.

#### Distance Reads Zero / Incorrect?

Inspect sensor wiring connections to Pin 9 and 10, and ensure your battery supply voltage is sufficient.

#### Safety & Practical Engineering Tips

- Always verify battery polarity and terminal connections before powering up the system!
- Use **Jugaad** (creative problem-solving skills) to adjust physical chassis balance and refine sensor angles!

# Student Practice Sheet

Test Your Knowledge — Autonomous Rover & Sensor Logic

Name: \_\_\_\_\_

Date: \_\_\_\_\_

- Q1.** What component serves as the "brain" of the obstacle-avoiding rover, and what is its primary role?
- Q2.** Describe how the HC-SR04 Ultrasonic Sensor measures the distance to an object in front of the rover.
- Q3.** Why is the total pulse duration divided by 2 in the mathematical formula for calculating distance?
- Q4.** Identify the default Arduino digital pins specified in the wiring guide for the Ultrasonic Trig pin and Echo pin.
- Q5.** What will happen to the rover's behavior if the threshold distance in code is set to 2 cm instead of 15 cm?
- Q6.** If your rover moves backward when it encounters an obstacle instead of stopping and turning right, what two steps can you take to fix it?
- Q7.** *Super Tinkerer Challenge:* Draw an obstacle course scenario and write the conditional decision steps (if/else) the rover should execute when approaching a wall.